Food Packaging

Functions & Requirements

- Packaging is the science, art, and technology of enclosing or protecting products for distribution, storage, sale, and use.

Definitions

- The *packaging institute international* defines packaging as a enclosure of products, items or packages in a wrapped pouch, bag, box, cup, tray, can, tube, bottle or other container form for any of the following functions

- containment,
- protection and /or preservation,
- communications and utility or
- performance.

- The *UK Institute of packaging* provides three definitions of packaging.

1. A coordinated system of preparing goods for transport, distribution, storage, retailing and end-use.
2. A means of ensuring safe delivery to the ultimate consumer in sound condition at minimum cost.
3. A techno-economic function aimed at minimizing cost of delivery while maximizing sales.

Difference between packing and packaging

Category	Packaging	Packing
Mean	It is a process of designing and manufacturing container for protection and also for display	It is building a container or box for a product for transport and storage but not for a display
Purpose	To contain and protect	To ease transportation
Food contact	Comes in direct contact with the product	Never in direct contact with the food
Material	Primary packaging	Secondary/tertiary/quaternary

Types/Levels of packaging

- **Primary packaging**, commonly referred to as *consumer* or *retail packaging*, is the box that is in direct contact with your product.

- **Secondary packaging** is the shipping box that is used to contain your primary packaging, but it can also be retail or consumer-facing packaging.

- **Tertiary packaging** is generally the shipping box used to protect the secondary packaging. Warehouses use tertiary packaging to protect and transport goods

Functions of packaging

Containment (to contain the product)

- The ability of the packaging to maintain its integrity during the handling of the food while processing, transportation, marketing and disposing.
- Without containment product loss would be wide spread.
- Depends on the product's physical form and nature
- The containment function of packaging contribute to the protection of the environment as faulty packaging could result in food waste and pollution

Protection (to protect the product)

- Packaging protects the food against biological contamination (microorganisms, insects, rodents), chemical changes (moisture, color, oxidation, flavor etc.) and mechanical damage during distribution and use

Preservation (to extend the shelf-life)

- Shelf life is defined as the time limit, under defined storage conditions, during which food remains safe, retains desired sensory characteristics

- It preserves the food by extending the shelf-life by providing a barrier between the food and the external environment

Communication (to provide the information)

- Provides information about the food product such as brand identification, pricing, legal requirements, product ingredients, shelf-life, and pricing.

- Helps to promote the food as it informs to consumers about many offers i.e. free extra product, new product, money off etc.

Convenience (to ease the access of food)

- Range of sizes, easy handling, easy opening and dispensing, resealability, and food preparation in the package are examples of packaging providing convenience to the consumer.

Requirements for effective food packaging

- Be non toxic
- Protect against contamination from microorganisms
- act as a barrier to moisture loss or gain and oxygen ingress
- protect against ingress of odors or environmental toxicants
- Filter out harmful UV light
- Provide resistance to physical damage
- Be transparent, tamper – resistant or tamper – evident
- Be easy to open
- Have dispensing and resealing features
- Be disposed of easily,
- Meet size, shape and weight requirements
- Have appearance, printability features
- Be low cost
- Be compatible with food